

VALIDITY OF GARMIN WRIST-WORN SMARTWATCH BIOMECHANICAL VARIABLES DURING OVERGROUND WALKING AND RUNNING WITH AND WITHOUT A PRAM

Tash Panucci¹, Danielle Trowell¹, Natalie Saunders¹, Claire Kenneally-Dabrowski¹, Monica Kelly¹

¹ Institute for Physical Activity and Nutrition, Deakin University

Email: npanucci@deakin.edu.au

INTRODUCTION

Wearable devices are increasingly used to monitor running biomechanics outside laboratory settings. However, the validity of commercially available wrist-worn devices varies according to the biomechanical variable, device configuration and testing conditions [1]. Wrist-worn devices derive running dynamics from upper-limb movement using accelerometry and proprietary algorithms, potentially making estimates sensitive to changes in arm movement [1, 2]. Multi-sensor approaches incorporate a chest-mounted accelerometer which may provide additional information about trunk movement and whole-body mechanics [3]. Running with a pram constrains natural arm swing and alters aspects of running mechanics. Thus, providing a relevant condition in which to examine whether wrist-derived estimates remain valid. Therefore, this study aimed to determine the validity of Garmin wrist-worn smartwatch-derived biomechanical variables, with and without a Garmin chest strap, compared with three-dimensional (3D) motion capture during overground walking and running with and without a pram.

METHODS

Nine recreationally active adults (7 females, 2 males; age 32.8 ± 3.1 years) completed overground walking and running trials. Participants completed walk, jog, run and fast-run conditions, with five trials per condition and up to three min rest between trials; condition order was counterbalanced. Pram trials were completed during walking and running. Two Garmin Forerunner 165 wrist-worn smartwatches were used to obtain biomechanical variables, with one device paired with a Garmin HRM-Pro chest strap. 3D kinematics were simultaneously recorded using a 24-camera Vicon motion capture system (200 Hz) with a full-body Plug-in-Gait marker set. Systems were synchronised using Unix timestamps. Garmin-derived cadence, stride length, ground contact time (GCT), and vertical oscillation (VO) were compared with corresponding motion-capture measures. Agreement was assessed using Bland-Altman analyses and Intraclass correlations ($ICC_{3,1}$). Statistical significance was accepted as $p < 0.05$.

RESULTS

Cadence demonstrated good–excellent agreement with Vicon ($ICC = 0.859–0.938$, $p < 0.001$), with excellent agreement observed across three of four conditions. Mean bias was small (0.78–5.90 steps/min), although proportional bias was observed during running without the pram (Figure 1).

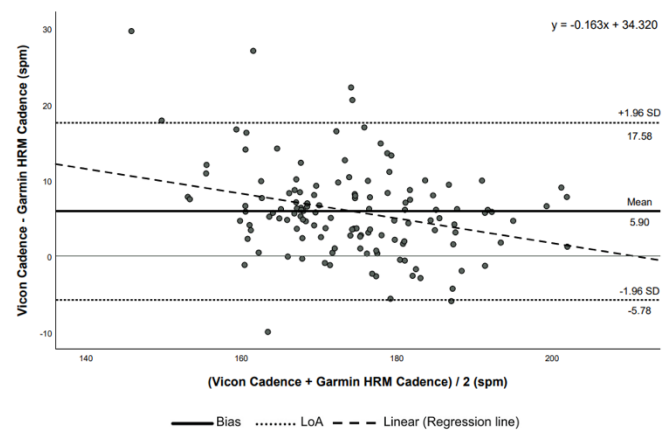


Figure 1: Bland-Altman analyses displaying the mean bias and limits of agreement (LoA) between the Vicon motion analysis system and Garmin HRM for cadence during running without the pram.

CONCLUSIONS

Garmin-derived cadence demonstrated good–excellent validity compared with 3D motion capture, supporting its use for field-based monitoring. Further analyses will determine validity across the additional biomechanical variables and the influence of pram running.

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REFERENCES

- [1] Mason R et al. Sports Med 53: 241-268, 2023.
- [2] Adams D et al. J Orthop Sports Phys Ther 46: 471-476, 2016.
- [3] Benson LC et al. Gait Posture 63: 124-138, 2018.

	Condition	ICC (95% CI; <i>p</i>)	Mean bias	ULoA	LLoA
Walk	No Pram	0.922 (0.862, 0.957; $p < 0.001$)	1.43	4.86	-2.00
Run	No Pram	0.859 (0.806, 0.898; $p < 0.001$)	5.90	17.58	-5.78
Walk	Pram	0.938 (0.889, 0.966; $p < 0.001$)	0.78	1.47	-2.10
Run	Pram	0.933 (0.881, 0.963; $p < 0.001$)	1.81	7.57	-3.95

ICC, Intraclass correlation; CI, Confidence interval; ULoA, Upper limit of agreement; LLoA, Lower limit of agreement